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8. Write a program to implement Linear Regression (LR) algorithm in python

1. House Price Prediction

Dataset: house.csv

Area (sq.ft)	Price (Lakhs)		
800	40		
1000	50		
1200	60		
1500	75		
1800	90		

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("house.csv")
X = df[["Area"]]
y = df["Price"]
model = LinearRegression()
model.fit(X, y)
sample = [[1400]]
print("Predicted Price:", round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 70.0 Lakhs

2. Salary Prediction

Dataset: salary.csv

Experience (Years)	Salary (Lakhs)
1	3
2	4
3	5
4	6
5	7

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("salary.csv")
X = df[["Experience"]]
y = df["Salary"]
model = LinearRegression()
model.fit(X, y)
sample = [[6]]
print("Predicted Salary:", round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Salary: 8.0 Lakhs

3.

Study Hours	Marks
2	40
3	50
4	60
5	70
6	80

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("marks.csv")
X = df[["Study Hours"]]
y = df["Marks"]
model = LinearRegression()
model.fit(X, y)
sample = [[7]]
print("Predicted Marks:", round(model.predict(sample)[0], 2))
```

OUTPUT

Predicted Marks: 90.0

4.

Advertising	Sales
10	25
20	35
30	45
40	55
50	65

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("sales.csv")
X = df[["Advertising"]]
```

```
y = df["Sales"]
model = LinearRegression()
model.fit(X, y)
sample = [[35]]
print("Predicted Sales:", round(model.predict(sample)[0], 2))
```

OUTPUT

Predicted Sales: 50.0

5.

Temperature	Sales
20	100
25	125
30	150
35	175
40	200

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("icecream.csv")
X = df[["Temperature"]]
y = df["Sales"]
model = LinearRegression()
model.fit(X, y)
```

```
sample = [[32]]
```

```
print("Predicted Sales:", round(model.predict(sample)[0], 2))
```

OUTPUT

Predicted Sales: 160.0